# Effects of Odor Familiarity on the Development of Systematic Exploration in the Spiny Mouse, *Acomys cahirinus*

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Developmental changes in patterns of exploration by infant spiny mice, *Acomys cahirinus* are described. These were investigated using an unbaited radial maze and under three odor conditions; that is, animals were tested in the presence of either familiar (own) odors, unfamiliar conspecific odors, or no odors (washed floor). Two hypotheses were tested. The first was based on the results of a pilot study, and was that adult animals would explore the radial maze in a systematic and predictable fashion tending to move from one arm to the next sequentially. It was hypothesized that infants would adopt this sequential strategy only gradually, as they matured. The second hypothesis was that such patterns of exploration would depend upon the olfactory environment; the presence of familiar odors might facilitate systematic patterns of exploration in very young *Acomys*. The results showed that both adults and juveniles were less likely to move sequentially when tested with no conspecific odor present; sequential patterns of movement were most likely in the presence of unfamiliar odor, however. The only significant change with age in infants was found for animals tested with unfamiliar odors; these animals showed a dramatic increase in sequential behavior between 3 and 7 days of age. Two additional experiments are reported, which investigated the preferences of infant *Acomys* for unfamiliar conspecific odors, and it was found that very young (about 3 days) animals exhibit a preference for odors derived from unfamiliar conspecific litters, even when tested in the physical presence of their own parents. The results are discussed with reference to the use of olfactory information as directional cues for animals exploring the radial maze.

Most mammals spend a great deal of time exploring or patrolling their environment, gaining information about the location of resources, or of possible danger. Despite this importance of exploration, we know surprisingly little about how free-living animals explore novel environments. The development of patterns of exploratory behavior in young animals has also been neglected. Yet exploration

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is an important part of development for any young animal; through exploration, the animal not only learns about its environment, but it probably also learns "how to learn" from its environment, acquiring skills relevant to information-gathering in general.

In this study, we investigated the development of systematic exploration in the precocial Spiny Mouse, *Acomys cahirinus*. This small, easily bred animal is the only murid rodent to have precocial young, and has been the subject of a number of studies of olfactory imprinting (e.g., Porter & Ruttle, 1975; Porter & Etscorn, 1976). Precocial young are faced with somewhat different problems than the young of altricial species: most murid young are born blind and relatively immobile, remaining in the nest for many days after birth. *Acomys* young, by contrast, are not in specific nests (Dieterlen, 1962) and are born sighted and mobile. It is thus important that they learn rapidly the characteristics of the mother and of the immediate environment, to avoid high predation risks.

We chose to study spiny mice because of their precocity. Since the young can move around independently from birth, we can thus study exploratory behavior and its development from the day of birth itself. Our study aimed primarily to investigate how patterns of systematic exploration develop in *Acomys*: that is, do young animals have to "learn to explore," to learn the rules of efficient exploration? Do patterns of exploratory activity become more efficient as the animal gains experience? [By "systematic," we mean exploration with a nonrandom, but efficient, pattern—such as a tendency to visit different areas sequentially and not to return to areas recently sampled.]

There is evidence that small mammals, both in the wild and in the laboratory, do explore systematically, even in the absence of obvious reward such as food (Barnett & Cowan, 1976; Cowan, 1982; Bättig et al., 1976). The first of the present experiments was thus designed to investigate whether very young *Acomys* explore in the systematic fashion of adult rodents, and if not, to determine at what age such systematic exploration might first appear.

A radial maze was used to study the extent to which patterns of movement during exploration are systematic. Radial mazes are often used in research on the working memory and spatial behaviour of animals (Olton et al., 1979): usually, the animal is set the task of obtaining food from each of several baited arms. The most efficient strategy is to visit each arm only once, and rats, for example, quickly learn to do this efficiently. Performance in the radial maze can then be used to assess working memory following, say, drug treatment or damage to the brain (e.g., Olton et al., 1979; Rauch & Raskin, 1984). But even in the absence of obvious reward in the arms, the frequency with which an animal enters each arm can measure systematic exploration. Barnett and Cowan (1976), for example, recorded the frequency of entries into arms in a residential cross maze as an index of "patrolling" behavior. Rats searching for food rewards in radial mazes do tend to behave systematically, often moving in a consistent clock- or counterclockwise fashion so that all arms are explored in order (Olton, 1978; Olton, Collison, & Werz, 1977)). Similarly, laboratory mice exploring an enclosed radial maze quickly adopt this sequential or "kinaesthetic" strategy (Pico & Davies, 1984).

Adult spiny mice also tended to employ this strategy in pilot trials in unbaited enclosed radial mazes, frequently turning into adjacent arms. This pattern did not appear to be simply the consequence of the mice hugging the walls of the maze, because the animals usually returned briefly to the open space of the central arena before making the next choice (unpublished observations). We therefore used the criterion of sequential turns to measure the extent to which infant *Acomys* explore the maze systematically. We do not know whether the maze provides an adequate test of the degree to which free exploration is systematic in more complex environments. The radial maze does, however, have the advantage that it constrains the movement of the animal sufficiently to make its behavior relatively definable.

These experiments utilized a form of radial maze, and investigated the ways in which the animals use the arms and how their visits were ordered. Although we focused upon young animals, and how exploration develops, the first experiment also included some adults, both to provide a comparison and to replicate and extend the findings of the pilot study. The first hypothesis to be tested was that (i) adults would tend to employ a sequential strategy when exploring the maze, even in the absence of obvious reward; and (ii) that this tendency would develop in young animals as they learned about their environment and developed relevant skills. We also attempted to discover the importance of olfaction in their exploratory behavior. These animals can presumably employ olfactory cues for orienting themselves: for a start, newborn spiny mice are known to learn rapidly the odor of conspecifics and of the natal area. The odor of the home area could thus enable a young spiny mouse to orient toward or away from the parents. Hence we also considered the possibility that the odor of conspecifics, including parents, might influence the way the animals explored the radial maze. Specifically, we tested the hypothesis that the presence of familiar (e.g., parental) odors would facilitate systematic exploration in young animals, perhaps by providing directional information by which the young animal could orient itself.

## Experiment 1: The Development of Systematic Exploration in the Radial Maze

### Methods

Forty-six infant *Acomys* were divided into three groups, as described below. We also tested 32 adults in the radial maze, in similar fashion to provide a comparison. Group 1 animals consisted of adult breeding pairs and their offspring to be tested in the presence of familiar (homecage) odors ($N$ = 10 adults; 11 infants). Group 2 animals consisted of adult pairs and young tested with unfamiliar odors (i.e., obtained from other breeding pairs with young: $N$ = 12 adults; 15 infants). The 3rd group consisted of similar adults and young tested in a clean maze ($N$ = 16 adults; 20 young). As litters were born a complete litter and parents were allocated at random to one of the three groups. Analyses of adult and infant behavior were, however, carried out separately.

All animals were housed in cages 25 × 40 cms, with pine shavings as bedding. They were fed rat chow and sunflower seeds, with water ad libitum, with a twice-weekly supplement of pieces of apple. Infants began testing on the 3rd day of life (day 3) and were subsequently retested on days 7, 10, 14, 17, and 21. They were not handled apart from being placed individually into the apparatus on those days. All animals were bred in the Animal House of the Open University, and were maintained on a reversed lights schedule (lights on from 2100 to 0900 hrs).

All behavioral tests took place between 0915 and 1500 hrs. For testing, each animal was removed from its homecage and introduced into the center of the apparatus. It was then observed for 10 min. The apparatus consisted of a six-arm radial maze, with the arms leading off a central hexagon, which was 40 cm across at its widest point Each arm was 30 cm long and 10 cm wide and could be entered from the hexagon by a small door 5 cm high. For animals in groups 1 and 2, the arena contained soiled bedding as an odor source, which was placed in the arena immediately before testing. If siblings from the same litter were being tested consecutively, the odor source was left undisturbed between trials; for non-siblings, it was removed after testing any one animal. The arena was then washed down, and a new odor source put down as appropriate. For group 3 animals, the arena was washed down with dilute detergent before and after testing each animal. Odor sources for group 1 and 2 animals were not spread evenly through the apparatus, but only in the hexagon and along one arm. This arm was chosen at random and rotated between tests.

The movements of the animal were observed using a video camera mounted immediately above the maze, and connected to a Panasonic videocassette recorder. Behavior was then transcribed from the videotape using an Apple II microcomputer.

For analysis, data were analyzed in terms of durations spent in different areas of the maze, and of the sequence of arm entries. Three measures of the tendency toward sequences were used, namely, the mean frequencies of: (i) repeats to the same arm (R); the number of entries made before all arms were sampled (V); and sequences of entries to adjacent arms. This consisted of a score (A) of the mean frequency of all visits to immediately adjacent arms, irrespective of the direction of turning, (e.g., going from arms 1–2, 4–5, 3–2 or 6–1) and a score (S) to represent the use of longer "runs" in the same direction of entries into adjacent arms (e.g., 1–2–3–4; or 2–1–6–5–4–3; or 5–6–1). *S* was calculated on the basis of the *total* number of sequential entries, however long each individual "run" of sequential entries was. Thus 3 runs of 3 consecutive adjacent turns would have the same weight in calculating *S* as 1 run of 9 consecutive adjacent turns.

## Results

The results of the experiment are shown in Figs. 1–4 and Table 1. Data for adults were compared across groups for each measure, for both sexes, using a two-way analysis of variance, while those for juveniles were compared across ages using a two-way analysis of variance with repeated measures (both sexes

*TABLE 1. Number of Arm Entries Taken before All Arms of Maze Visited (Mean ± s.e.m.).*

| | Group | | |
|---|---|---|---|
| | 1 | 2 | 3 |
| 1. Adults | | | |
| Males | 7.6 ± 1.6 | 11.2 ± 3.8 | 13.0 ± 1.4 |
| Females | 8.6 ± 0.8 | 9.4 ± 1.1 | 10.5 ± 1.5 |
| 2. Juveniles | | | |
| Day 3 | 17.9 ± 3.2 | 16.7 ± 2.1 | 16.9 ± 2.3 |
| 7 | 16.7 ± 2.2 | 12.1 ± 1.1 | 13.5 ± 1.6 |
| 10 | 14.4 ± 1.5 | 10.5 ± 1.0 | 10.9 ± 1.3 |
| 14 | 9.7 ± 1.1 | 11.4 ± 1.5 | 13.5 ± 1.1 |
| 17 | 11.5 ± 1.7 | 9.8 ± 1.1 | 10.8 ± 1.2 |
| 21 | 8.2 ± 0.6 | 13.1 ± 1.8 | 13.7 ± 1.7 |

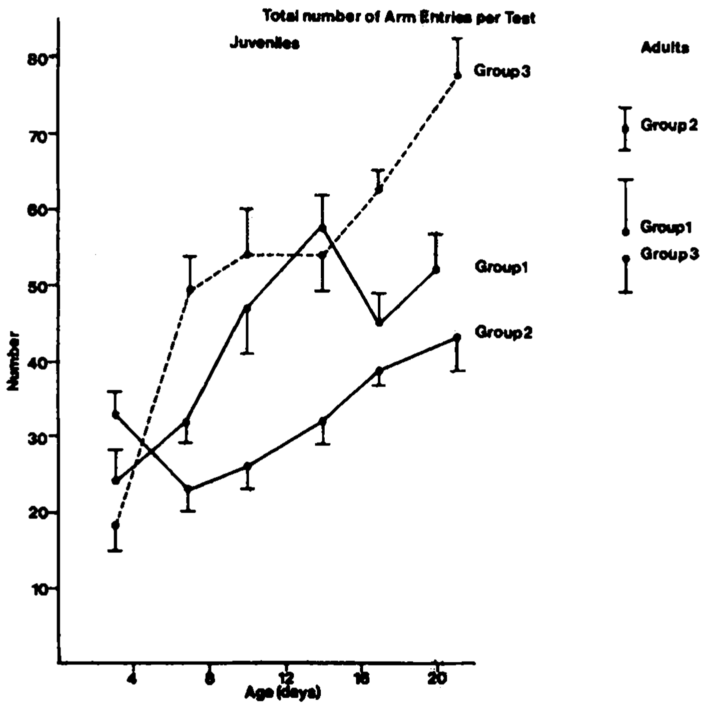


Fig. 1. The total number of visits to each arm of the radial maze during tests by infants of different ages, and by adults in each group (means ± s.e.m.).

combined). Table 1 shows the results for $V$, the number of arm entries taken before all arms of the maze were visited. For adults, there were no significant differences on this measure; for juveniles, however, there were significant effects due to age ($F_{(5,251)} = 11.8$; $p < 0.01$), but no significant effect due to group ($F = 2.3$; n.s.).

Figure 1 shows the total number of visits ($N$) to any arm. For adults, there were no significant differences on this measure, but for juveniles, there was a significant effect due to group ($F_{(2,43)} = 18.9$; $p < 0.01$), and of age ($F_{(5,251)} = 28.5$; $p < 0.01$), as well as a significant age × group interaction ($F_{(5,251)} = 7.2$; $p < 0.01$).

Figure 2 shows the mean duration of time (in seconds; calculated from totals for each animal) that animals spent in the odorant arm (or equivalent in the case of group 3). There were significant differences between groups for adults ($F_{(2,43)} = 38.7$; $p < 0.01$), with group 2 animals spending considerably more time in the odorant arm. [Concomitantly, there were significant intergroup differences for time spent in central hexagon and in other arms; data not shown.] Similar differences were found for juveniles: there was a significant difference in time spent in odorant arm ($F_{(2,43)} = 4.91$; $p < 0.05$).

Figures 3 and 4 show the results for the two measures of sequential tendency ($A$ and $S$, expressed as percentages of all arm entries). There were significant differences due to group for adults on both measures: for adjacent turns ($A$), $F_{(2,43)} = 11.1$; $p < 0.01$), and for longer sequences ($S$), ($F_{(2,43)} = 11.0$; $p < 0.01$). For juveniles, the analysis of $A$ showed signficant effects due to group ($F_{(2,43)} = 7.0$; $p < 0.01$), and age ($F_{(5,251)} = 6.0$; $p < 0.01$), and a significant age × group interaction ($F_{(5,251)} = 3.7$; $p < 0.01$). Sequential entries ($S$) showed only an effect due to age

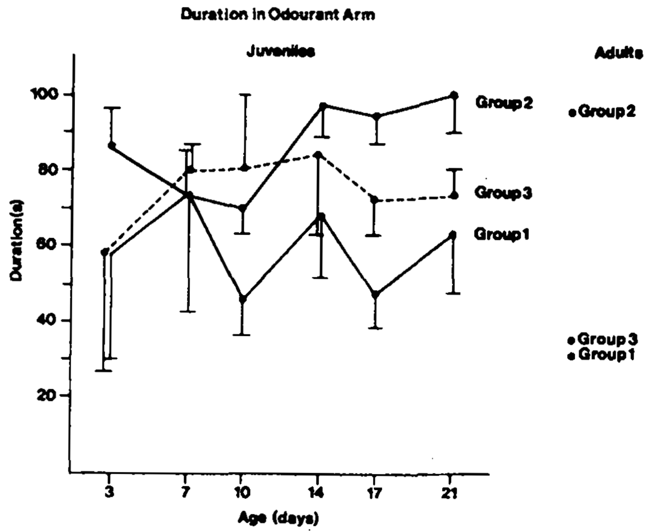


Fig. 2. The total duration time (seconds) that animals spent in each arm of the radial maze.

($F_{(5,251)}$ = 12.1; $p < 0.01$). However, this analysis was based on the absolute number of sequential entries, and when the measure of *S* was calculated relative to total entries (*N*; as it is in the percentages shown in Fig. 4), then it is clear that there are differences between the groups; those animals tested with unfamiliar odors present do not show differences from other groups in the raw scores, but they did show fewer overall arm entries. Thus, the relative scores for these animals (*S*/*N*) suggest that they show a higher than average use of the sequential strategy.

## Discussion

The first hypothesis tested in this experiment was that adults in the radial maze would consistently move from one arm to the next adjacent, while infant

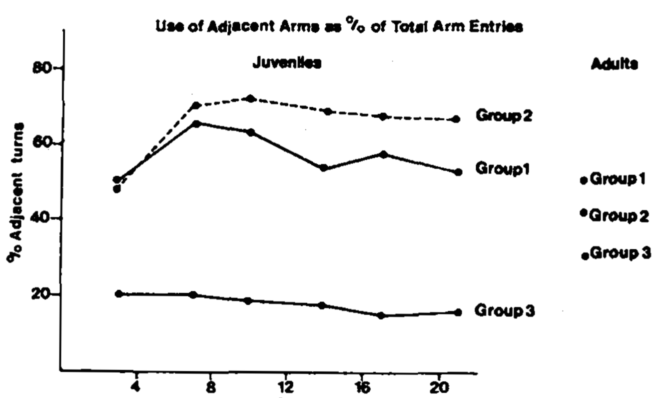


Fig. 3. The percentage of all entries that involve turns from one arm to an immediately adjacent arm, for infants at each age and for adults.

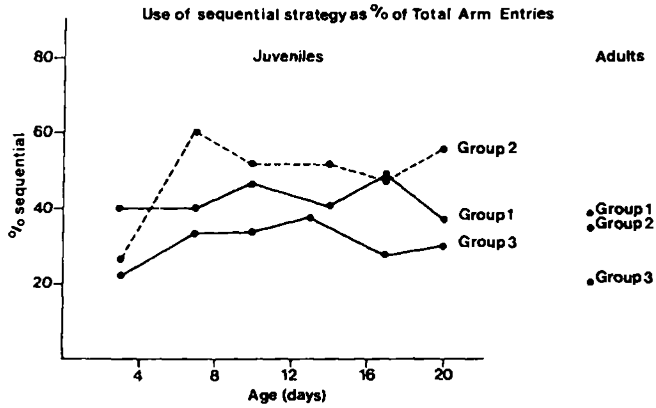


Fig. 4. The percentage of all entries that involve sequences of adjacent turns; that is, sequences of three or more successive turns into adjacent arms (for explanation see text).

*Acomys* would exhibit such "sequential" behavior only gradually, as they matured. Adults did tend to use a sequential strategy. Groups differed, however, on measures of sequencing. Those animals tested on a clean floor used the sequential strategy significantly less often than those tested with either familiar or unfamiliar odors present. In the absence of conspecific odor, for example, only about 20% of all arm entries formed part of a sequence of 3 consecutive adjacent turns or more, compared to approximately 37% of turns for animals tested in the presence of odor. These results suggest that, for adults at least, olfactory information helps the animal to move systematically while exploring.

The hypothesis is thus partially supported for adult animals. That is, they do sometimes use a sequential strategy, but rather more often when olfactory information is available in the radial maze. For juveniles, the hypothesis was that the ability to use a sequential strategy would mature gradually. Although there were differences between the groups analogous to those shown by adults, infant spiny mice appeared to be quite capable of exploring the maze in a sequential fashion even from day 3.

Differences between the groups were, however, stronger for juveniles. Thus, infants tested on a clean floor used the sequential strategy less often than did either of the other groups. The groups also differed in their activity: on day 3, animals tested on the clean floor were rather less active than others (as shown by the number of arms they entered), although at other ages, the reverse was true. Animals in group 2 (unfamiliar odor) were the least active.

Juveniles in group 2 also most often explored in a sequential way, as defined particularly by the measure $S$. On average, 48% of their turns were in sequential runs, compared to 42% for group 1 and 30% for group 3. Moreover, group 2 uniquely showed an effect of age on the use of the sequential strategy: at day 3, only 26% of turns were in sequences, whereas by day 7, 60% were.

The latter was, in fact, the only age difference in sequential strategy found in this experiment. That is, from day 3 at least, infant spiny mice are quite capable of adopting a sequential strategy to explore the radial maze systematically. Whatever skills are required to adopt sequential behavior, such as the ability to remember where one has just been, perhaps, or to remember always to turn in one particular direction, are skills that spiny mice appear to have even by the 3rd day of life. They are, however, more likely to demonstrate these skills in the presence of conspecific odors.

We also analyzed the distribution of time spent in different parts of the maze. Adults spent more time in the odorant arm if it contained unfamiliar conspecific odors. This is perhaps not surprising; what is more surprising is that similar differences emerge for very young animals. Juveniles tested in the presence of familiar odors of the homecage spent less time than animals of other groups in the areas containing that odor; this was particularly true on day 3. Three-day-old animals from groups 2 and 3, for example, spent approximately 40–45% of their time in the central (odorant) area, compared to 25–30% spent by group 1 animals. This appeared to result from the animal becoming immobile in a particular location after an initial period of exploring all the arms. Although all animals did sample each arm at least once, it is not clear that this immobility indicates a preference for unfamiliar odors *per se* by very young infants, or perhaps a tendency not to become immobile in the presence of familiar odors, simply because the animal was not making a choice *between* odors but only between areas containing or not containing conspecific odors. The next two experiments attempted to address the problem of odor preference, by presenting young spiny mice with a choice between several different conspecific odors in order to determine their preferences in the radial maze.

## Experiment 2: Preferences for Different Conspecific Odors by Infant *Acomys*

### Methods

Fifteen infant spiny mice were used in this experiment, housed and maintained as described for experiment 1. Testing was carried out as described above, in the radial maze, and the movements of the animals recorded using the microcomputer. For this experiment, each arm of the radial maze contained an odor source. These were in constant serial order, but were rotated between tests and between each test for any one animal so that the location of each odor was not in any constant position within the room. The odor sources were: (i) soiled bedding from the animal's own homecage; (ii) soiled bedding from another litter with young of approximately similar age; (iii) clean shavings; (iv) soiled bedding from a cage containing only adult males; (v) soiled bedding from adult females; and (vi) soiled bedding from a breeding pair currently without young. As before, each animal was put into the apparatus and its behavior observed for 10 min. after which it was returned to its homecage. Testing took place on days 3, 7, 10, 14, 17, and 21 as before.

Because this experiment was concerned specifically with the animal's preference for particular odors, the duration of time spent in any part of the maze was measured. No attempt was made in this experiment to measure sequences in detail as was done in the first experiment.

### Results

The results of this experiment are summarised in Table 2. Data were analyzed using a two-way analysis of variance with repeated measures for the age variable. There was no significant effect of age alone, but there was a significant effect of

*TABLE 2. Preferences for Conspecific Odors in a Radial Maze (mean duration ± s.e.m.).*

| Odor source | Age 3 | Age 7 | Age 10 | Age 14 | Age 17 | Age 21 |
|---|---|---|---|---|---|---|
| Parental | 76.5 ± 23.6 | 77.5 ± 6.9 | 73.6 ± 9.6 | 83.3 ± 10.9 | 80.9 ± 9.9 | 54.0 ± 7.5 |
| Unfamiliar parental | 195.6 ± 55.0 | 66.3 ± 18.0 | 79.7 ± 8.3 | 89.5 ± 9.2 | 92.0 ± 23.1 | 89.4 ± 30.0 |
| Clean bedding | 27.3 ± 7.5 | 42.2 ± 5.7 | 90.1 ± 22.1 | 94.1 ± 10.0 | 71.7 ± 9.0 | 66.7 ± 12.2 |
| Adult males | 73.5 ± 35.1 | 99.5 ± 11.5 | 96.6 ± 19.8 | 63.9 ± 6.5 | 85.2 ± 12.9 | 67.1 ± 13.5 |
| Adult females | 89.8 ± 36.0 | 104.6 ± 20.0 | 95.2 ± 22.1 | 71.1 ± 9.9 | 88.0 ± 13.9 | 44.7 ± 8.2 |
| Breeding pair | 42.5 ± 10.0 | 88.1 ± 26.3 | 88.4 ± 7.7 | 93.3 ± 10.2 | 81.4 ± 14.2 | 175.5 ± 56.3 |
| Central hexagon | 103.3 ± 43.0 | 123.0 ± 9.3 | 78.8 ± 10.1 | 107.3 ± 8.9 | 104.1 ± 8.2 | 107.7 ± 20.8 |

the location of the odor ($F_{(6,98)} = 2.8$; $p < 0.05$), and a significant age × odor location interaction ($F_{(30,629)} = 2.19$; $p < 0.01$). The data shown in Table 2 suggest that this was largely due to marked differences in time spent in different arms by animals at day 3 and day 21.

## Discussion

The results of this odor-choice experiment suggest that at most ages, young spiny mice do not distinguish much between odors. For most of the ages tested, there is little difference between the times spent in particular arms. There are, however, marked differences at day 3 and day 21. At day 3, animals spent longer in the arm containing unfamiliar odor from another litter, while at day 21, they spent longer in the arm containing odor from the breeding pair without young.

That there should be such a difference at day 21 seems at first sight surprising. Inspection of the raw data, however, suggested that this was largely due to three animals, which for some reason spent most of their time in that area. None of the other 21-day-olds exhibited such a preference. If these data are analyzed separately, then the significant difference disappears. The data for day 3 animals, by contrast, do show a significant effect even when analyzed separately, and 12 out of 15 animals showed a preference for the arm containing odor from another litter.

The finding that, at 3 days of age, there is a preference for the odor of an unfamiliar litter supports the suggestion from our first experiment that young spiny mice do prefer such an odor to that of their own, familiar homecage odor. This preference does not appear to be due to novelty *per se,* because all the conspecific odors other than that from the homecage would be novel in this experiment. In addition, day 3 animals, unlike older ones, seem to avoid the arm containing shavings alone, spending considerably less time in that arm than any other.

It is not clear why such young animals should prefer the odor of another litter. One possibility is that the familiar homecage odor is not attractive in the physical absence of the parents, although this would only explain why they do not find that odor attractive, but would not explain why they find the other litter odor attractive. It is at least likely that odor from another litter would have some similarity to odors from the pup's own litter.

In the 3rd experiment, we considered the possibility that the physical presence of the parents might alter the preferences shown by very young animals. This focuses specifically on the 1st week of life, when animals might still be learning about the olfactory characteristics of their environment. We gave the infants a choice in the radial maze between nonodorant arms, an arm containing soiled bedding from a conspecific litter as before, and an arm containing both soiled homecage bedding and the parents themselves, contained in a small metal cage.

# Experiment 3: Preferences of Infant *Acomys* in the presence of the Parents

## Methods

Twenty-eight infant spiny mice were used in this experiment, housed and maintained as previously described. Testing was carried out as before, in the radial maze, except that one arm of the radial maze was replaced by a short tunnel leading to a mesh cage, approximately 10 × 15 × 20 cm, into which the parents were placed. The location of this arm was varied randomly between tests. Prior to testing, the parents were placed into the cage, and a small quantity of soiled bedding from the homecage was placed into the entrance to the tunnel. A similar quantity of soiled bedding from an unfamiliar conspecific litter was also placed in the entrance of the arm immediately opposite the one containing the parents. All other arms remained empty. At the end of the test, the infant being tested was returned to the homecage and then the next animal from that litter (if any) was tested; infants were thus not kept from their parents for any longer than 50 min (in the case of litters of 4).

Because this experiment was concerned specifically with preferences in very young animals, and the effect on these of the parents presence, tests were carried out on days 1, 3, 5, 7, and 9. As before, the times spent in each arm was recorded, and the time spent in the central hexagon. No measures of sequencing were recorded in this experiment.

## Results

The results of this experiment are presented in Table 3, and were analysed using a two-way analysis of variance with repeated measures, as before, to compare times spent in different parts of the maze. Data are presented as time spent near parents (that is, in the entrance to the tunnel); time spent in the opposite, odorant arm; time spent in either of the arms adjacent to the parents; and time spent in the remaining two arms. These analyses showed significant effects of location (that is, which arm was chosen), for which $F_{(3,76)} = 7.04$; $p < 0.01$). There was no significant effect of age ($F = 0.73$; n.s.), although there was a significant age × location interaction ($F_{(12,304)} = 3.9$; $p < 0.01$).

## Discussion

The data shown in Table 3 indicate that very young animals do indeed prefer to spend their time in the vicinity of the unfamiliar conspecific odor, even when homecage odor is combined with the physical presence of the parents. At days 1 and 3, there is a preference for the unfamiliar odor. By day 5, however, this tendency has switched to a slight preference for remaining near the parents, and, at days 7 and 9, to a marked preference for the parents.

*TABLE 3. Preferences for Parents vs. Conspecific Odor (Experiment 3).*

| | Odor source/Arms visited | | | |
|---|---|---|---|---|
| Age (days) | Parents | Conspecific odor | Arms adjacent to parents | Arms adjacent to odor |
| 1 | 49.4 ± 27.0 | 158.0 ± 57.0 | 121.0 ± 50.0 | 81.4 ± 39.0 |
| 3 | 115.6 ± 40.0 | 179.7 ± 47.0 | 67.6 ± 24.0 | 131.6 ± 42.0 |
| 5 | 197.8 ± 38.0 | 139.8 ± 20.0 | 83.8 ± 14.5 | 98.0 ± 19.5 |
| 7 | 223.5 ± 35.0 | 115.1 ± 17.4 | 110.9 ± 14.3 | 73.9 ± 11.1 |
| 9 | 261.4 ± 26.0 | 83.9 ± 12.2 | 120.4 ± 12.3 | 77.3 ± 8.9 |

It is, of course, possible that very young animals actually avoided the parents, because the parents rarely remained still in the cage, but typically turned somersaults and were very active. Older infants would have experienced this type of parental activity, but it is possible that very young animals found the noise thus generated aversive and remained away from the source. Certainly, being removed from the homecage and parents and placed into an unfamiliar environment is likely to be highly aversive to a young infant, although that would not necessarily explain why they seem to *avoid* being near the parents and to remain elsewhere in the apparatus.

Whether or not these young animals do find the activity of the parents aversive, there does appear to be a preference for the arm containing the conspecific unfamiliar odor, even in day-old animals. This finding supports the suggestion from the previous 2 experiments that infant spiny mice actually prefer to explore or remain near odors obtained from other litters. It would appear that the actual presence of the parents makes little difference to this choice, except in somewhat older animals.

## General Discussion

These experiments sought to test two hypotheses, namely that (a) infant spiny mice will show only gradually the ability to move *systematically* around the radial maze as they acquire the skills that enable them to "learn to explore" (rather than, say, moving randomly or moving according to a consistent rule of turning); and (b) that the presence of familiar odors would facilitate systematic exploration in young animals, perhaps by providing a discriminative cue by which the infant could locate itself.

The 1st of these predictions was upheld only partially. Indeed, if a strategy of sequential and adjacent turns is a measure of systematic exploration then infant spiny mice can explore the radial maze systematically even by the 3rd day of life. This ability is, however, marked only when the animals are tested in the presence of conspecific odors. On a clean floor, animals make more adjacent turns than expected by chance, but these turns are rarely ordered into longer sequences. In the presence of odors, in contrast, animals tend to make sequences of adjacent turns. Sometimes these sequences are quite long, even more than 20 consecutive adjacent turns in the same direction in the case of one 17-day-old animal.

Yet even in the absence of odors the movements of the animals are not random. Animals might generate such patterns because they can remember where they have recently explored, or because they adopt a consistent rule, such as continuing to move clockwise. They might also create such patterns by moving along the walls of the apparatus. Such "wall-hugging" behavior would automatically lead the animal to keep going around the apparatus in the same direction. We did not, however, observe obvious wall-hugging. Animals of all ages tended to reenter the arena and move a few centimeters into the central hexagon, clear of the walls. In the case of very young animals, they would then remain there for a few seconds before moving off into another arm; older animals would usually move almost immediately into the next arm. Moreover, even if wall-hugging did occur at times, it would not explain the differences between groups tested in different olfactory conditions.

The 2nd prediction does, however, seem to be upheld: in the presence of odors, young animals (as well as adults) are more likely to behave systematically in the radial maze. The present experiments do not indicate how this olfactory information might be used; they do indicate that conspecific odors can differ in their effects on exploration. Specifically, very young animals, about 3 days old, prefer to remain near odors derived from breeding pairs with young. From day 7 onward, the presence of such odors appears to facilitate the use of a sequential strategy for exploring the maze more than does the presence of familiar homecage odor.

Experiments 2 and 3 demonstrated the existence of this preference for the odors of unfamiliar litters. It appears to be a preference specifically for odors from this source, rather than unfamiliar conspecific odors *per se*, and appears—in very young animals at least—even to occur in the physical presence of the parents. A preference for odors from conspecific litters was reported by Porter and Ruttle (1975), who concluded that very young *Acomys* have developed specific attachments to chemical stimuli produced by specific classes of conspecifics (that is, other litters). They also suggest that such attachments would encourage "highly mobile infants to restrict their movements to an area frequented by other infantile and maternal conspecifics," which would, they suggest, be highly adaptive, because lactating spiny mice have been reported to nurse infants indiscriminately (Dieterlen, 1962).

As the infant gets older, it might also learn the specific characteristics of its parents and approach them directly even in unfamiliar settings such as a radial maze. What is not clear from these results, however, is why, at the youngest ages, infants appear to be oblivious to the odors derived from their own homecage: if odors from breeding adults with young are generally attractive, why is the familiar odor not similarly attractive? Certainly, Porter and Ruttle (1975) found no such discrimination; in their study, very young spiny mice did not differentiate between homecage odor and that from another litter. We obtained a clear preference for odors from other litters in these experiments, although these results do not, unfortunately, indicate why our animals differed from those tested in Porter and Ruttle's study.

Somewhat older infants appear to be less dependent upon the presence of familiar odor; that is, they do not remain specifically in areas containing familiar (or slightly unfamiliar) odor sources. Yet at all ages, exploration is more systematic in the presence of conspecific odor than on a newly washed floor. The presence of such odors might provide the animals with directional information. An animal might, for example, use the odorant area as a reference point from which to begin (or by which to remember) sequences. This, however, does not seem likely from the data obtained here, because we found no evidence of sequences tending to begin with that arm more than any other. A second possibility is simply that, as the animal moves about, it spreads some of the odorant bedding, leaving faint trails by which it might subsequently orient. The present experiments do not allow us to evaluate this possibility.

A 3rd, and perhaps more interesting, possibility is that odor in the one arm provides a reference point within the maze that enables the animal to orient itself spatially with respect to extramaze cues, which might then be used to provide secondary reference points. Experiments using radial mazes with rats have often suggested that animals do indeed use extramaze cues for spatial orientation (e.g., Olton, 1978). On the other hand, these experiments often use an elevated, open maze, which the rats have to traverse in order to obtain food from cups at the ends of the arms. Such a maze is rich in spatially relevant cues. Pico and Davies (1984), in experiments using laboratory mice, suggested that only the use of the elevated, open maze truly allows the animal to utilize spatial cues from outside the maze. Schenk (1985) has recently demonstrated that, in rats, only animals older than 40 days are able to demonstrate adultlike navigation using spatial cues; by contrast, rats younger than about 25 days are neither able to use spatial cues for navigation (Rauch & Raskin, 1984), nor show spontaneous alternation (Douglas et al., 1973). Whether the precocial spiny mouse shows similar age-related changes in navigational ability, and whether any such changes are specific to open or closed mazes is not yet known. Young spiny mice develop rapidly in the first week of life and it remains to be determined how their abilities to orient and explore the environment develop in detail.

## Notes

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